Prediction of Microwave-Assisted Double-Bond Transfer by Machine Learning

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Abstract: This paper proposes an algorithm that models molecules as graphs, and uses graph kernel and kernel machine to predict the performance in rearranging Schiff base derived from furfural and diphenylamine catalysed by different solvents under microwave irradiation (exclusive of benzoic acid dissolved in methylbenzene). Several kernel functions are tested, and it is concluded that the ODD-STh kernel is particularly suitable for the task.

Keywords: machine learning, prediction, double-bond transfer, kernel method, graph kernel

INTRODUCTION

Machine learning simulates the functions of human learning, including features such as the discovery of rules from a large amount of data, the extraction of relevant knowledge, and constant improvement through practice. The ultimate goal of the

intelligent machine can only be achieved by making computer systems capable of human learning. Machine learning forms the core of artificial intelligence research and has been applied in a wide range of fields. As an experimental science, chemistry has been studied for a long time, and a vast amount of knowledge has been accumulated in the process. Faced with the complexity of chemical systems, computers have been used to deal effectively with chemical problems, so that chemistry has achieved unprecedented development. However, completing a chemical experiment not only takes time, but also bears an economic burden and contributes to environmental pollution. This has motivated researchers to study the use of machine learning to replace chemical reaction experiments.[1-3]

Double-bond transfer is a common chemical reaction, usually catalysed by alkali[4-8] or acid.[9-11] Since alkali or acid can cause environmental pollution, it is necessary to conduct research on green double-bond transfer. This paper investigates double-bond transfer of the Schiff base derived from furfural and diphenylamine catalysed by different solvents under the assistance of microwave irradiation. Based on the experimental results, a prediction model of microwave-assisted double-bond transfer is established by machine learning.

RESULTS AND DISCUSSION

Experiments were carried out on the double-bond transfer of Schiff base **1**, catalysed by solvents using microwave irradiation, which results in the formation of **2**, as shown in Figure 1.

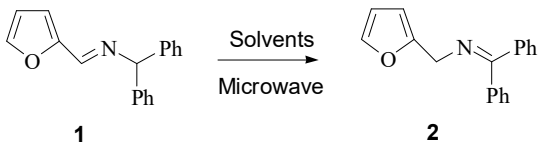
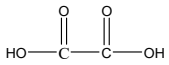
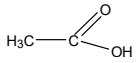
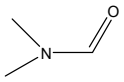
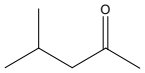
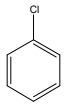
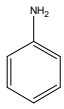
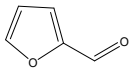
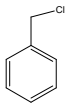
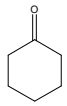
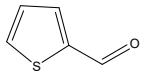
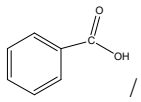
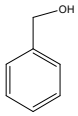
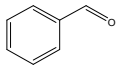


Figure 1. Double-bond transfer of Schiff base **1**

Table 1. Double-bond transfer of Schiff base **1** using solvents and microwave irradiation (640w, 60min)

Entry	Reagents	Yield (%)
1		45%
2		4%
3		5%
4		5%
5		6%
6		90%
7		98%
8		95%

9		92%
10		95%
11	 / methylbenzene	94%
12		40%
13		40%
14		97%

The predictive capability of the program was evaluated using data from 14 small molecular substances from our experiments, as shown in Table 1. The experimentally observed results were compared to the program predictions. The results showed a 67% accuracy rating in predicting the experimentally observed rearrangements of the Schiff base under microwave irradiation, which was considered to be very acceptable. In particular, the gap in predictivity was mainly caused by not considering the influence of reaction temperature and functional groups. The program has provided a valuable method to help guide further experiments.

EXPERIMENTAL

Molecule modelling

In order to predict the performance of molecules with machine learning algorithms, the first step is to find a mathematical model for molecules. A graph is a natural choice. The formal definition of a graph is:

Definition 1:

A graph is a sextuple $G = (V, E, V_L, V_A, E_L, E_A)$, where:

- V is the set of nodes;
- E is the set of edges;
- V_L is the set of node labels, and V_A is the set of node attributes. Node labels and attributes both contain supplementary node information. By convention, node labels are categorial data while node attributes are numerical data;
- E_L is the set of edge labels, and E_A is the set of edge attributes. Similar to node labels and attributes, edge labels are usually categorial data and edge attributes are usually numerical data.

To model a molecule as a graph, we can take atoms as nodes and chemical bonds as edges. Further, types of atom (for example, carbon or oxygen) can be set as node labels, and chemical bond type (such as single or double-bond) can be set as edge labels. Figure 2 shows the graph of formic acid.

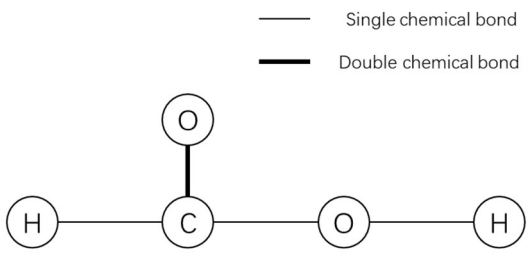


Figure 2. Graph of formic acid

Kernel method and graph kernel

After building the mathematical model of a molecule, the next step is to apply an appropriate prediction algorithm. Using the kernel method with graph kernel is one choice. In a machine learning context, similar input objects (i.e., all molecules) are called kernels, and a kernel function is used to compute kernel. The idea behind the kernel method is that similar objects should have similar properties. In our task, this means that molecules with a similar structure should exhibit a similar performance. After computing the kernel, we can use kernel-based machine learning algorithms, also known as kernel machines, to predict performance. Support vector machine (SVM) is the most successful kernel machine and is used in our experiment.

In recent years, a great number of kernel functions have been developed, based on different concepts, such as random walks and subgraphs. In our experiment, we tested many different kernel functions, and found that ODD-STh kernel is both theoretically suitable to our task and practically powerful. Introduced by Giovanni Da San Martino when computing the similarity of two molecule graphs,[12] the ODD-STh kernel function first decomposes the two graphs to ordered directed acyclic graphs (DAGs) as subgraphs, and takes the sum of similar DAGs as the similarity of two molecule graphs, as follows:

$$K_{K_{DAG}}(G_1, G_2) = \sum_{\substack{D_1 \in DD(G_1) \\ D_2 \in DD(G_2)}} K_{DAG}(D_1, D_2) \qquad (1)$$

In Formula 1, $DD(G_i)$ corresponds to a graph decomposition of graph G and K_{DAG} is

the kernel function of DAGs. To compute K_{DAG}, the DAGs are again decomposed into groups of subtrees, and kernel is calculated as the sum of tree kernels between derived trees from each of the two DAGs, as follows:

$$K_{DAG} = \sum_{\substack{v_1 \in V(D_1) \\ v_2 \in V(D_2)}} C\big(T(v_1), T(v_2)\big) \tag{2}$$

In Formula 2, $\mathrm{T}(\cdot)$ corresponds to the tree-visits on DAGs, and $\mathrm{C}(\cdot)$ is the tree kernel introduced by Vishwanathan [13] As shown in Figure 3, the whole formic acid molecule is decomposed into smaller trees in the right column including the function groups (carboxyl), which have a strong relationship with the performance of the molecule and makes ODD-STh suitable for our task. Further, the ODD-STh kernel can handle node labels, and provided us with the highest prediction accuracy in our experiment.

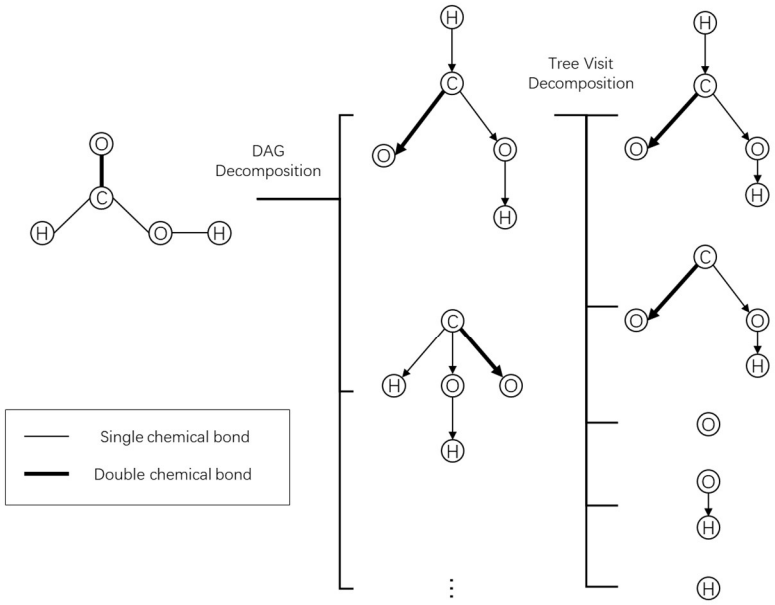


Figure 3. Molecular graph decomposition process

Graph kernel is not the only method of graph-based machine learning. The graph

mining method transforms a graph into a vector by extracting its structural statistical characters,[14] such as average node degree or edge weight. However, it is difficult for methods based on graph mining to extract local structure, and also to utilize node label information. Methods based on graph embedding first embed graphs into equivalent vectors and use normal vector-based machine algorithms.[15] Inspired by the success of convolutional neural networks (CNNs), graph convolutional networks (GCNs) have attracted increasing attention in recent years. The last two methods are both reasonable methods, and will be tested in future work.

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